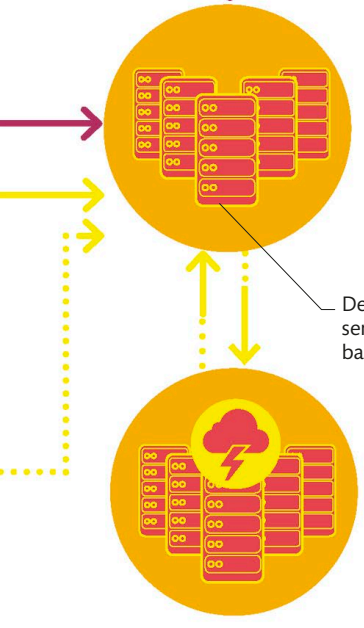


5 Smartphone app
The heating request is sent to another digital device—in this case, the user's smartphone, which operates a smart heating app. The app controls the thermostat in the house and sends a signal back to the smart speaker to indicate that the user's request has been completed.



4 4 Device service provider
This software recognizes requests and directs them to the appropriate service, which may be another server in the cloud. The Paris request will be sent to a weather database. The heating request will be directed to an app on the user's smartphone.

Device service provider sends weather information back to smart speaker

5 Weather database
The device service provider accesses the weather database to find the forecast temperature and chance of rain in Paris. The data travels to the smart speaker via the device service provider.

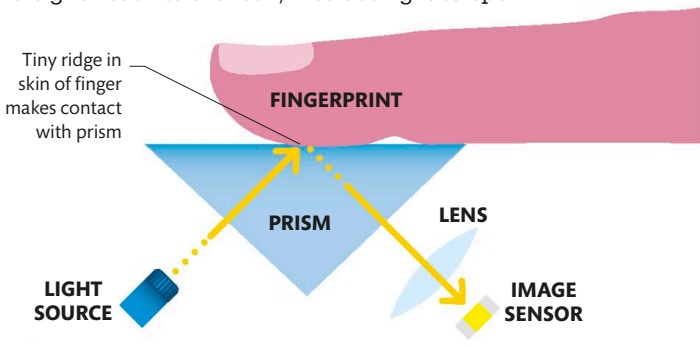
THE INTERNET OF THINGS

Billions of devices embedded with microprocessors and communications technology can connect to the Internet, communicate with other machines or humans, and share data, such as through machine-readable QR (quick response) codes. This network of devices is known as the Internet of Things.

QR CODE

Biometric locks

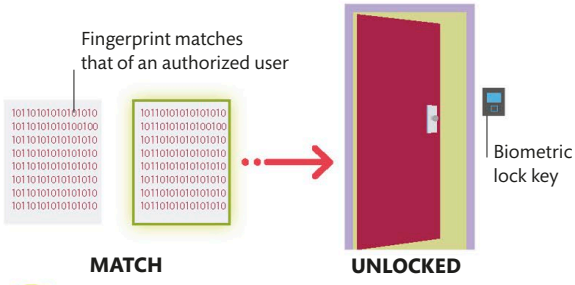
Increasing numbers of digital devices, such as electronic door locks, replace physical keys with scanners. These capture a biometric feature of a person, such as the iris pattern or fingerprint. Software distills this image down to a unique pattern stored in a database. A match triggers a signal back to the lock, instructing it to open.



1 Optical fingerprint scanner
LED light travels through a prism, bounces off the finger placed on the scanner, and is focused by a lens onto a digital image sensor such as a CCD chip. The sensor records the patterns of ridges and valleys that make up the fingerprint.



2 Analysis and algorithms
Software analyzes the fingerprint image, seeking out identifying features, such as joining lines (known as minutiae). The software uses an algorithm to create a digital template of the fingerprint.



3 Search and compare
The scanned pattern is sent to a database for comparison. If a match with an authorized user is found, an electronic signal will be sent back to the lock, instructing it to open to admit the person.